

BallPlate Simulation Exercise#1 v221107

Dear Master's Program Students in the KF Course,

enclosed is the basic MatLab program 'BallPlateSim0.m' plus a *.csv data file. Please make yourself familiar with the code:

The program reads the logfile 'BallPlateLogData.csv' and plots central variables. The file contains real-time log data from a run of the REAL ballplate system (all command & control input values, estimated state variables plus raw measurement data).

In part I of this exercise you should do the following:

- 1) Implement the continuous 2nd order system model (A_2, b_2) and derive the corresponding time discrete formulation (F_2, g_2). The values of F_2 and g_2 for the time-discrete system should be verified by at least one ALTERNATIVE formula / calculation approach !

NOTE: Higher n-th-order system models should later use (A_n, b_n) and (F_n, g_n, h_n) respectively.

- 2) Test the OPEN LOOP system responses to a sequence of steps of the servo actuator angle 'xThetServo', starting from 0° : $+30^\circ$ @ $t=1.0s$, -60° @ $t=2.0s$ and $+30^\circ$ @ $t=3.0s$, so that xThetServo is 0° again after the step sequence. Plot the time histories of the simulated ball position 'xPosSim' in [pxl] & the simulated ball speed 'xDotSim' in [pxl/s].

In part II of this exercise you should then solve the following tasks:

- 1) Compute the gain vector h_2 of a 2nd order observer for the observer poles at $s_0 = -10$ and $s_1 = -20$. The values of h_2 should be verified by at least one alternative formula / alternative calculation approach !
- 2) Implement & run the observer from real given xThetPWMcmd- and xPosMeas-values from the log file and plot the time histories of xPosSim2Obs and xDotSim2Obs in comparison to xPosMeas/xPosEstLog and xDotDifLog/xDotEstLog respectively.

The servo actuator angle command is a PWM signal with a pulse width between 1.0ms and 2.0ms :

Pulsewidth 1.5 ms -> neutral position, i.e. xThetServo = 0°
Pulsewidth 1.0 ms -> min. position, i.e. xThetServo = $-\text{THETA_SERVO_MAX}$ [$^\circ$]
Pulsewidth 2.0 ms -> max. position, i.e. xThetServo = $+\text{THETA_SERVO_MAX}$ [$^\circ$]
with $\text{THETA_SERVO_MAX} = 50^\circ$

An introduction to the Matlab program and some further hints will be given in the lecture on Monday Nov. 7th. In the future your Matlab code & results should

be submitted via a GitLab project directory (access tba.)

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